MTH647 - Quiz No.3

Date: 08 July 2025

Q.1 Which of the following best describes the effect of unit step inputs u(t−1) and u(t−2) on the system response?

A. They produce instantaneous jumps in velocity.

B. They turn the system on gradually over time.

C. They apply constant forces starting at t=1 and t=2.

D. They act as impulse forces at t=1and t=2.

Q.2 Given: f(t)=(t−3)2□u(t−3) What is the Laplace transform L{f(t)}?

A. 2/ s3

B. e−3s. 2/s3

C. 2e3s / s3

D. e-s s-2

Q.3 Which signal is represented by this diagram: zero for t < 0, then jumps to 1 at t = 0 and remains at 1?

A. Unit impulse

B. Unit parabola

C. Unit ramp

D. Unit step

Q.4 In solving the differential equation : y''+3y'+2y=δ(t−1), y(0)=y'(0)=0 what is the Laplace transform of the right-hand side?

A. e-s

B. se-s

C. δ(s−1)

D. 1/s

Q.5 Which of the following correctly represents the convolution of two continuous-time signals x(t) and h(t)?

A. y(t) = _{0}^{t} x(-1) \, h(1-) \, d

B. y(t) = _{-}^{} x() \, h() \, d

C. y(t) = _{0}^{t} x() \, h(t-) \, d

D. y(t) = _{0}^{} x() \, h(1-) \, d

Q.6 A voltage waveform starts at 0 V, jumps to 1 V at t = 1 s, to 2 V at t = 2 s, to 3 V at t = 3 s, and returns to 0 V at t = 4 s. Which of the following represents this signal using unit steps?

A. u(t−1)+u(t−2)+u(t−3)−3u(t−4)

B. u(t−1)+u(t−2)+u(t−3)

C. u(t−1)−u(t−2)−u(t−3)+3 u(t−4)

D. u(t−1)+2u(t−2)+3u(t−3)

Q.7 Which of the following statements correctly defines the Dirac delta function δ(t)?

A. It is a constant function with value 1 at t=0.

B. It is zero for all t, including at t=0.

C. It is infinite at t=0, zero elsewhere, and integrates to 1.

D. It increases linearly from 0 to 1.

Q.8 Which of the following is the correct Laplace transform of δ(t−3)?

A. e3s /s
B. s-3
C. e3s
D. e-3s

Q.9 Given L{t2}=2/s3 , what is the Laplace transform of (t−2)2u(t−2) using t-shifting theorem?

A. 2/s3. e2s
B. 2/(s3-2)
C. 2/s3. e-s
D. 2/s3. e-2s

Q.10 Which of the following correctly describes the relationship between the Dirac delta function δ(t) and the unit step function u(t)?

A. (t) = {d}{dt} u(t)
B. δ(t)=u(t)t
C. \ u(t)= {d}{dt} (t)
D. u(t)=δ(t) t
